Lines and Minds: Visual Abstraction in Art, Psychology, and Computer Graphics

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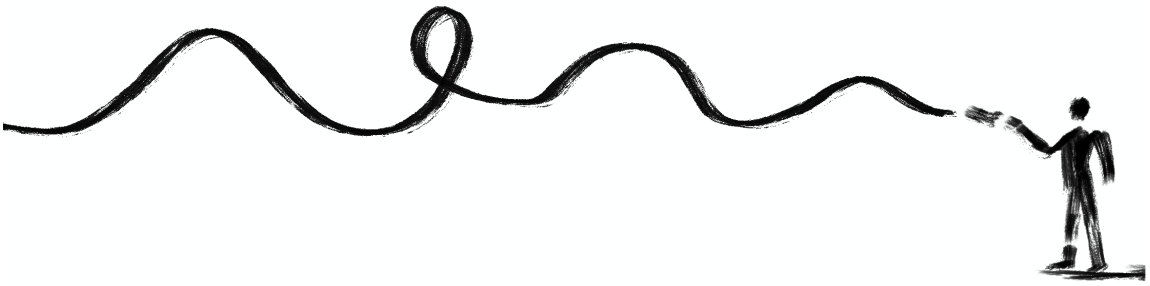

Visual abstraction — from sketches and diagrams to icons and comics — enables people to communicate, express, and create by distilling complex visual information into simplified forms that preserve essential meaning. These simplified forms support perception, reasoning, and communication in ways that detailed depiction often cannot. Visual abstraction has been studied across cognitive science, art theory, and computer graphics, offering complementary perspectives on how humans perceive, produce, and interpret such representations. This workshop brings together these perspectives to examine abstraction as a cognitive mechanism, a communicative tool, and a computational challenge, exploring questions such as: Why and how do people abstract? What makes a simplification effective? How can machines learn to abstract in ways aligned with human perception? And as AI excels at generating realistic images, what role might abstraction play in how humans and AI create together?

1 Motivation and Background

Visual abstraction is the process of simplifying or distilling visual information to its essential elements, reducing complexity while preserving, or even amplifying, meaning [Arnheim 1969; Viola and Isenberg 2018]. For example, in the famous "Le Taureau" series (Figure 1), Picasso progressively transforms a detailed anatomical rendering of a bull into a few essential lines that still capture the *essence* of the animal.

The ability to create and understand visual abstraction traces back to humanity's earliest forms of expression, such as cave paintings and hieroglyphics, used to convey experiences, tell stories, and symbolize beliefs [Clottes 2008; Gombrich

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1960]. This has persisted into modern times, where people across disciplines continue to rely on simplified visual forms such as sketches, diagrams, icons, maps, and more to communicate, learn, and create [Fan et al. 2023; Tversky 2011]. Given its breadth, visual abstraction has been studied across a range of disciplines: cognitive science [Fan et al. 2023; Mukherjee et al. 2023; Tversky 2002, 2011], psychology of art [Arnheim 1969; Chamberlain et al. 2015; Kozbelt 2001], art and design [Gombrich 1960; McCloud 1993], as well as computer graphics and vision [DeCarlo et al. 2003; Hertzmann 2003, 2021; Vinker et al. 2022a,b; Winnemöller et al. 2012], and human-computer interaction [Igarashi et al. 1999; Kazi et al. 2014]. We believe these fields can mutually inform and enrich one another. Insights from cognitive science and visual theory can inform the design of computational tools that align with how humans perceive and produce abstractions. Conversely, computational approaches offer new methods for testing theories and building systems that support human visual thinking. This workshop aims to build bridges across these communities and inspire new research at their intersection. We will explore questions such as: Why and how do people abstract? What makes an abstraction effective? How can computational systems learn to abstract in ways aligned with human perception? And as AI becomes increasingly capable of generating realistic images, what role might abstraction play in shaping more intuitive and human-centered tools for creation?

Fig. 1. "Le Taureau" by Pablo Picasso — note how the abstraction process is achieved by gradually *removing* elements while the bull's essence is preserved. ©"Pasadena, Norton Simon Museum, Picasso P. The Bull, 1946" photo by Vahe Martirosyan, [CC BY-SA 2.0] via (https://bit.ly/3MFB3pm).

By bringing together researchers from cognitive science, the visual arts, and computer graphics, we aim to deepen the conversation and open new research directions at the intersection of human and machine visual abstraction.

This workshop is part of a continued effort to build a community around these topics, following the COGGRAPH workshop at CogSci 2024 [Chandra et al. 2024] and the Drawing and Sketching workshop at SIGGRAPH 2025 [Vinker et al. 2025]. Last year's Drawing and Sketching workshop attracted strong interest within the SIGGRAPH community, with attendance exceeding room capacity. This response reflects a growing appetite within our community for these topics, and we anticipate continued enthusiasm for this year's workshop.

2 Objectives and Topics

This workshop brings together researchers and practitioners from computer graphics, cognitive science, and the visual arts to explore the nature and power of visual abstraction — how simplified representations amplify meaning, support thought, and enable communication. Our goal is to examine abstraction as a cognitive mechanism, a communicative tool, and a computational challenge, and to foster dialogue across disciplines that have studied these questions in parallel. We hope to surface shared principles, spark new collaborations, and consider how these insights apply as AI plays an increasing role in visual creation.

Topics of discussion will include:

The Psychology of Abstraction Why do people abstract, and how? We will explore how humans perceive, produce, and use simplified visual representations — examining the cognitive mechanisms that make abstraction effective, the choices artists and designers make when translating the world into visual form, and how expertise shapes these processes. What do artists see that others don't? How do we decide what to keep and what to leave out? And what role does the viewer play in completing meaning from sparse visual information?

The Language of Abstraction Lines, dots, arrows, boxes, gestures: how do simple visual elements carry meaning? We will examine abstraction as a form of communication — from the mark systems of diagrams to the visual vocabulary of comics — asking whether there are shared principles underlying how we read and make pictures. What makes a simplified image legible? What role does the viewer play in completing meaning from sparse visual information?

Computational Visual Abstraction. How can computational systems understand and generate abstract visual representations? We will explore the challenges of designing tools that interpret simplified input, produce meaningful visual output, and support human visual thinking. What can computational approaches reveal about how abstraction works — and what remains difficult for machines that comes naturally to humans?

Abstraction and Human-AI Collaboration Simplified visuals — sketches, gestures, diagrams — offer natural ways to communicate intent. As AI systems become central to visual creation and increasingly fluent in generating realistic images, what role might abstraction play? We will examine how abstraction can serve as an interface between humans and machines, and discuss how principles of abstraction could enhance AI — supporting clearer communication, interpretability, and more human-aligned outputs.

This interdisciplinary discussion aims to open up new directions for research on visual abstraction in computer graphics and AI, while inviting broader perspectives from cognitive science, art, and visual narrative. By bringing these fields together, we hope to explore how abstraction shapes the way we think, communicate, and create — and what role it might play as humans and machines increasingly collaborate.

3 Format and Activities

The workshop will run as a half-day session, structured around three invited talks and a moderated panel discussion. The session will open with a brief introduction framing the key questions the workshop will explore around visual abstraction and communication: why and how do people abstract, what makes abstraction effective, and how do these processes play out across disciplines and tools?

Three invited speakers will then each give a 40-minute talk, spanning computational approaches to depiction and stylization, the cognitive foundations of visual abstraction, and sketch-based interfaces and tools for visual expression. Together, the talks will bridge perspectives from computer graphics, cognitive science, and human-computer interaction.

After a short break, the session will close with a 50-minute moderated panel discussion, bringing together the three speakers and an additional panelist to reflect on connections across the talks, engage with audience questions, and identify promising directions for future research and collaboration.

The session will close with an open discussion, inviting attendees to reflect on connections across the talks, share perspectives from their own disciplines, and identify promising directions for future research and collaboration.

4 Target Audience

This workshop is for anyone interested in visual abstraction, including sketches, drawings, diagrams, icons, comics, and other simplified visual forms, whether as a subject of scientific study, a creative practice, or a computational challenge. This includes researchers in graphics and AI, artists, designers, and cognitive scientists. It will also appeal to anyone interested in joining a growing interdisciplinary community and discussing the future of technology in this context. By bringing these perspectives together, the workshop adds to SIGGRAPH's long-standing interest in visual expression, and invites the community to look at abstraction as both a technical challenge and a shared human–machine language.

5 Invited Speakers and Panelists

The invited speakers represent an interdisciplinary group of researchers spanning cognitive science, computer graphics, and human-computer interaction. Together, their talks will trace an arc from how people perceive and reason through visual abstraction, to how computational models can capture and replicate these processes, to how interactive tools can empower new forms of abstract visual expression.

- **Barbara Tversky (Stanford University / Columbia University)**
  Barbara Tversky is Professor Emerita of Psychology at Stanford University and Professor of Psychology at Columbia Teachers College. Her research examines spatial thinking, diagrammatic reasoning, and gesture, exploring how people use simplified visual representations to think and communicate. Her book *Mind in Motion: How Action Shapes Thought* articulates a broad theory of how spatial cognition underlies language and abstract thought.
- **Aaron Hertzmann (Adobe Research)**
  Aaron Hertzmann is a Principal Scientist at Adobe Research and an Affiliate Professor at the University of Washington. His work examines the intersection of computer graphics, machine learning, and art. He pioneered key techniques in stylization and rendering, and his research explores how people perceive, depict, and interpret visual representations.
- **Rubaiat Habib (Adobe Research)**
  Rubaiat Habib is a Senior Research Scientist at Adobe Research. His work lies at the intersection of human-computer interaction and computer graphics, focusing on sketch-based interfaces, dynamic drawings, and tools that make visual expression more accessible. His research has led to products such as SketchBook Motion (Apple's Best iPad App 2016) and contributions to Adobe Character Animator (Engineering Emmy 2019).

Following the talks, all three speakers will join a moderated panel discussion alongside Scott McCloud, whose work on visual narrative and comics theory brings a complementary artistic and theoretical lens to the conversation. The panel will explore the tensions and connections between these viewpoints, and invite audience participation to identify open questions and future directions.

- **Scott McCloud (Cartoonist and Comics Theorist)**
  Scott McCloud is a cartoonist and theorist best known for *Understanding Comics: The Invisible Art*, a foundational work on the theory of visual narrative. His analysis of iconic abstraction, closure, and the "amplification through simplification" that comics achieve has influenced fields from illustration to interface design. His work offers a rich framework for understanding how simplified images communicate meaning.

6 Organizers

Yael Vinker (MIT)

Yael Vinker is a Postdoctoral Associate at MIT CSAIL, working with Prof. Antonio Torralba at the intersection of computer graphics, visual design, and AI. Her research focuses on developing computational tools that model and support visual thinking, visual communication, and creative expression. She has made key contributions to generative sketching and visual abstraction, advancing the use of large pretrained vision-language models in this domain.

Mia Tang (Stanford University)

Mia Tang is a Ph.D. student at Stanford University, working with Professor Maneesh Agrawala on generative models for visual expression. Her research focuses on interactive, controllable AI systems, including those that use sketches to support creative tasks such as generation and exploration.

Kartik Chandra (MIT)

Kartik Chandra is a PhD student at MIT, advised by Prof. Jonathan Ragan-Kelley and Prof. Josh Tenenbaum. His work connects big ideas from computer graphics and cognitive science to understand the foundations of non-photorealistic depiction.

Maneesh Agrawala (Stanford University)

Maneesh Agrawala is a Professor of Computer Science and Director of the Brown Institute at Stanford. His research focuses on computer graphics, human-computer interaction, and visualization, leveraging cognitive design principles to enhance the effectiveness of visual communication tools.

Judith E. Fan (Stanford University)

Judith E. Fan is an Assistant Professor of Psychology at Stanford University. She directs the Cognitive Tools Lab, where she studies how people use external representations, such as drawings, to support learning, communication, and problem-solving. Her work offers deep insights into the cognitive and psychological foundations of sketching.

Aaron Hertzmann (Adobe)

Aaron Hertzmann is a Principal Scientist at Adobe Research and an Affiliate Professor at the University of Washington. His work examines the intersection of computer graphics, machine learning, and art. He pioneered key techniques in stylization and rendering, and his research explores how people perceive, depict, and interpret visual representations.

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